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unun-vna 0.6.0

unun-vna is a command-line tool for measuring the **small-signal efficiency** of HF UnUn transformers with an S-A-A-2 / NanoVNA V2 using the **one-transformer method**.

The tool is generalized:

- no nominal load/series resistor is assumed;
- the actual series-load fixture is measured as a complex impedance versus frequency;
- transformer efficiency is calculated from measured S11/S21 and the measured fixture;
- transformer impedance ratio is configurable;
- 1:49 remains the default only for convenience.

The lower-impedance side of the transformer is connected to Port 1. The higher-impedance side is connected to the series fixture and then Port 2.

1. Transformer ratio

--ratio means the **impedance** ratio

$$N = \frac{Z_{\text{high}}}{Z_{\text{low}}}.$$

These are equivalent:

--ratio 49
--ratio 1:49
--ratio 1/49

All mean N = 49.

Examples:

1:4 -> N=4, turns ratio 1:2
1:9 -> N=9, turns ratio 1:3
1:16 -> N=16, turns ratio 1:4
1:49 -> N=49, turns ratio 1:7
1:64 -> N=64, turns ratio 1:8

The default is N=49.

Ratios below 1 are rejected because this measurement topology expects the low-impedance side at Port 1. Reverse the transformer connection instead.

2. Generic measurement topology

Port 1 ---- UnUn ---- measured series fixture ---- Port 2
Z0 1:N Z0

Z0 is normally 50 Ω.

The fixture may have any resistance value. It is not inferred from the transformer ratio and it is not assumed to equal its nominal/DC value.

It is measured as

$$Z_s(f) = R_s(f) + jX_s(f).$$

3. Ideal matching load for a selected ratio

For an ideal transformer with ratio N, the target total high-side resistance for a perfect resistive match is

$$R_{\text{high,target}} = N Z_0.$$

Port 2 itself contributes Z0, so the ideal purely resistive series fixture would be

$$R_{\text{series,target}} = (N - 1) Z_0.$$

At Z0=50 Ω:

UnUn	N	Total high-side target	Ideal series fixture
1:4	4	200 Ω	150 Ω
1:9	9	450 Ω	400 Ω
1:16	16	800 Ω	750 Ω
1:25	25	1250 Ω	1200 Ω
1:49	49	2450 Ω	2400 Ω
1:64	64	3200 Ω	3150 Ω

These numbers are **diagnostics only**. They are never substituted for the measured fixture impedance.

4. Fixture characterization

For a pure series impedance between two equal Z_0 ports:

$$S_{11} = \frac{Z_s}{2Z_0 + Z_s}$$

$$S_{21} = \frac{2Z_0}{2Z_0 + Z_s}.$$

Therefore:

$$Z_{s,1} = 2Z_0 \frac{S_{11}}{S_{21}}$$

and

$$Z_{s,2} = 2Z_0 \left(\frac{1}{S_{21}} - 1 \right).$$

A pure series element also satisfies

$$S_{11} + S_{21} = 1.$$

The fixture command reports both impedance estimates, their disagreement, and $|S_{11} + S_{21} - 1|$.

This checks whether the physical fixture can safely be represented by one complex series impedance.

5. Efficiency does not depend on the configured ratio

By S-parameter definition,

$$|S_{21}|^2$$

is the incident Port-1 power fraction delivered to the matched Port-2 termination Z_0 .

The same current flows through the Port-2 resistance and the series fixture. Thus total real high-side load power is

$$\frac{P_{\text{high}}}{P_{\text{incident}}} = |S_{21}|^2 \left(1 + \frac{\Re\{Z_s\}}{Z_0} \right).$$

Power accepted at Port 1 is

$$\frac{P_{\text{accepted}}}{P_{\text{incident}}} = 1 - |S_{11}|^2.$$

Therefore

$$\eta = \frac{|S_{21}|^2 \left(1 + \frac{\Re\{Z_s\}}{Z_0} \right)}{1 - |S_{11}|^2}$$

and

$$L_{dB} = -10 \log_{10}(\eta).$$

VSWR

The analysis also reports Port-1 VSWR derived directly from calibrated S11. With

$$\rho = |S_{11}|,$$

$$\text{VSWR} = \frac{1 + \rho}{1 - \rho}$$

for $\rho < 1$.

A perfect match gives VSWR = 1.0. Total reflection gives infinite VSWR. Values with $|S_{11}| \geq 1$, which can arise from residual calibration or measurement error, are reported as infinite rather than as a non-physical negative VSWR.

The tool also reports the **ideal lossless VSWR** for the configured transformer ratio with the actually measured fixture. This helps separate load mismatch from transformer loss.

There is no N in the measured efficiency equation. That is intentional.

The measured power flow already contains the transformer's actual behaviour; the ratio is not needed to reconstruct efficiency.

6. What --ratio is used for

The configured ratio is used to calculate a **lossless ideal reference** for the actual measured fixture.

The real measured high-side load is

$$Z_{\text{load}} = Z_s + Z_0.$$

An ideal lossless transformer with impedance ratio N reflects that to Port 1 as

$$Z_{\text{in}} = \frac{Z_{\text{load}}}{N}.$$

The input reflection expected from load mismatch alone is

$$\Gamma_{\text{ideal}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}.$$

Therefore

$$|S_{11,\text{ideal}}| = |\Gamma_{\text{ideal}}|.$$

The accepted incident power of that ideal network is

$$1 - |\Gamma_{\text{ideal}}|^2.$$

Since an ideal transformer is lossless, the fraction of that real power dissipated in Port 2 is

$$\frac{Z_0}{Z_0 + \Re\{Z_s\}}.$$

Hence

$$|S_{21,\text{ideal}}|^2 = (1 - |\Gamma_{\text{ideal}}|^2) \frac{Z_0}{Z_0 + \Re\{Z_s\}}.$$

The analysis output shows measured S11/S21 beside these ideal values.

This is useful when the fixture is not exactly the perfect matching resistance for the selected ratio.

7. S-A-A-2 enhanced-response calibration

The host-side S21 calibration follows the NanoVNA V2 **enhanced-response** user-calibration algorithm used for the on-screen traces. This is important for the one-transformer method because the series-load fixture is normally highly reflective.

7.1 Reflection calibration

Port-1 S11 uses the normal three-term SOL error model

$$m = e_{00} + \frac{e_t \Gamma}{1 - e_{11} \Gamma},$$

where e_{00} is directivity, e_t is reflection tracking and e_{11} is source match. SHORT, OPEN and LOAD determine these terms.

7.2 Transmission leakage

The NanoVNA V2 firmware models residual forward leakage as an affine function of the raw reflection measurement:

$$L(m_{11}) = L_0 + L_r m_{11}.$$

The two leakage states are taken from **SHORT and OPEN S21**. Both standards ideally have zero true transmission, so their measured S21 is treated as feed-through/leakage.

7.3 THRU reflection is part of the calibration

The THRU capture now stores both raw S11 and raw S21. DUT and THRU leakage are corrected with their own reflection states:

$$T_{\text{DUT}} = m_{21,\text{DUT}} - L(m_{11,\text{DUT}}),$$

$$T_{\text{THRU}} = m_{21,\text{THRU}} - L(m_{11,\text{THRU}}).$$

The THRU is therefore not assumed to have exactly zero reflection.

7.4 Source-match loop-gain correction

After SOL calibration, the NanoVNA V2 enhanced-response algorithm calculates

$$G(\Gamma) = \frac{1}{1 - e_{11}\Gamma}.$$

For the DUT and THRU this gives

$$G_{\text{DUT}} = \frac{1}{1 - e_{11}S_{11,\text{DUT}}},$$
$$G_{\text{THRU}} = \frac{1}{1 - e_{11}S_{11,\text{THRU}}}.$$

The calibrated forward transmission is then

$$S_{21} = \frac{T_{\text{DUT}}}{T_{\text{THRU}} G_{\text{DUT}}/G_{\text{THRU}}}$$

or equivalently

$$S_{21} = \frac{T_{\text{DUT}}}{T_{\text{THRU}}} \frac{1 - e_{11}S_{11,\text{DUT}}}{1 - e_{11}S_{11,\text{THRU}}}.$$

For a well-matched DUT this correction is small. For a fixture with $|S_{11}|$ close to one it can be large enough to matter by more than a dB.

8. S-A-A-2 Port-2 load-match limitation

The S-A-A-2 is a T/R VNA, not a full bidirectional two-port VNA.

It measures S_{11} and forward S_{21} , but cannot perform a complete independent Port-2 load-match correction.

High-ratio transformer/fixture combinations can be strongly reflecting when viewed from Port 2 even when Port 1 is well matched.

A THRU verification can therefore look excellent while a highly reflective DUT still has biased S_{21} .

Recommended Port-2 pad

Use a fixed 50 Ω attenuator, typically 10 dB, immediately before Port 2:

Port 2 ---- 10 dB pad ---- calibration plane ---- DUT

The same pad must remain in place during:

1. calibration;
2. fixture characterization;
3. transformer measurement.

The THRU calibration removes its forward loss and phase. Reflections from Port 2 pass through it twice, so a 10 dB pad reduces the load-match influence by roughly 20 dB.

9. Calibration

```
unun-vna calibrate \  
  --start 1M \  
  --stop 35M \  
  --points 501 \  
  -o hf35-pad
```

The program requests:

1. SHORT on Port 1 - both S11 and S21 are retained;
2. OPEN on Port 1 - both S11 and S21 are retained;
3. 50 Ω LOAD on Port 1 - used for the SOL reflection model;
4. THRU between the calibration planes - both S11 and S21 are retained.

SHORT/OPEN S21 define the leakage model. THRU S11 is required for the source-match enhanced-response correction.

Files:

```
hf35-pad.npz  
hf35-pad.npz.json  
hf35-pad.npz.through.csv
```

10. Characterize a fixture

The fixture extraction itself is independent of ratio.

For a 1:9 reference:

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 1:9 \  
  -o fixture-1to9
```

For 1:64:

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 64 \  
  -o fixture-1to64
```

--ratio changes only the target-load diagnostics printed by this command.

11. Measure a transformer

Measurement acquisition itself requires no ratio:

```
unun-vna measure \  
  --cal hf35-pad.npz \  
  -o transformer
```

The measured chain is:

Port 1 ---- transformer ---- characterized fixture ---- Port 2 pad ---- Port 2

12. Analyze

Default 1:49:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  -o efficiency.csv
```

Explicit 1:49:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  --ratio 1:49 \  
  -o efficiency.csv
```

1:9:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture-1to9.npz \  
  --ratio 1:9 \  
  -o efficiency-1to9.csv
```

1:64:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture-1to64.npz \  
  --ratio 64 \  
  -o efficiency-1to64.csv
```

13. Analysis CSV

The analysis CSV contains:

```
frequency_hz  
s11_db  
vswr  
s21_db  
fixture_r_ohm  
fixture_x_ohm  
impedance_ratio_high_over_low  
target_high_side_resistance_ohm  
target_series_resistance_ohm  
ideal_lossless_s11_db_for_measured_fixture  
ideal_lossless_vswr_for_measured_fixture  
ideal_lossless_s21_db_for_measured_fixture  
accepted_fraction  
delivered_fraction  
efficiency  
efficiency_percent  
transformer_loss_db
```

The target_* and ideal_lossless_* columns depend on --ratio.

The actual efficiency columns do not.

14. Ideal sanity checks

1:9

At $Z_0=50\ \Omega$, use an ideal $400\ \Omega$ series fixture:

$$400 + 50 = 450 \, \Omega$$

and

$$450/9 = 50 \, \Omega.$$

Thus an ideal transformer has

$$S_{11} = 0$$

and

$$|S_{21}| = 1/3$$

or approximately

$$S_{21} = -9.54 \, \text{dB}.$$

Efficiency evaluates to 100%.

1:49

With 2400 Ω series:

$$|S_{21}| = 1/7$$

or

$$S_{21} \approx -16.90 \, \text{dB}.$$

1:64

With 3150 Ω series:

$$|S_{21}| = 1/8$$

or

$$S_{21} \approx -18.06 \, \text{dB}.$$

15. Remaining assumptions

The tool does **not** assume:

- a 1:49 transformer;
- a 2400 Ω fixture;
- any nominal fixture resistance;
- that the fixture is purely resistive.

It does assume:

1. the lower-impedance transformer side is connected to Port 1;
2. the fixture is adequately represented by one complex series impedance;
3. Port 2 is a matched real Z_0 termination at the calibrated reference plane;

4. calibration, fixture measurement and DUT measurement use the same RF setup;
5. a fixed Port-2 pad, if used, remains in place;
6. the measurement is small-signal.

A fully arbitrary two-port fixture would require S_{12} and S_{22} and therefore a full bidirectional VNA.

16. Full workflow example

```
pipx install --editable . --force
```

```
unun-vna info
```

```
unun-vna calibrate \  
  --start 1M \  
  --stop 35M \  
  --points 501 \  
  -o hf35-pad
```

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 1:49 \  
  -o fixture
```

```
unun-vna measure \  
  --cal hf35-pad.npz \  
  -o transformer
```

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  --ratio 1:49 \  
  -o efficiency.csv
```

17. Source structure

```
unun-vna/  
├── pyproject.toml  
├── README.md  
├── src/  
│   └── unun_vna/  
│       ├── __init__.py  
│       ├── cli.py  
│       └── analysis.py  
└── tests/  
    └── test_math.py
```

`analysis.py` deliberately separates:

- ratio-independent measured efficiency;
- ratio-dependent ideal-reference diagnostics.

This prevents a nominal transformer ratio from silently changing the measured power calculation.